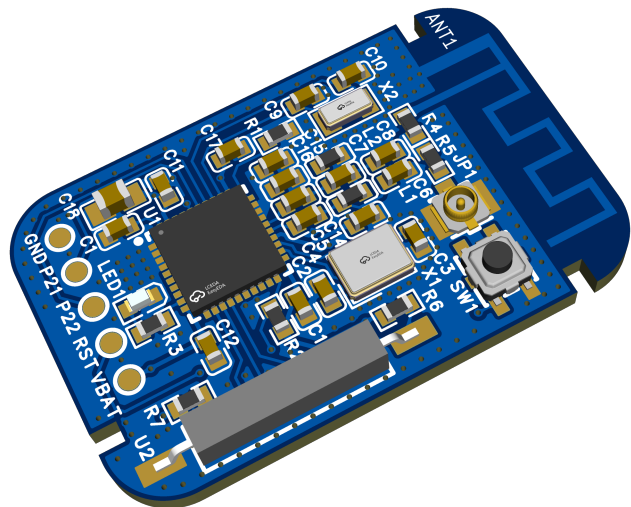




Zigbee End Device V2 (Contact Sensor)

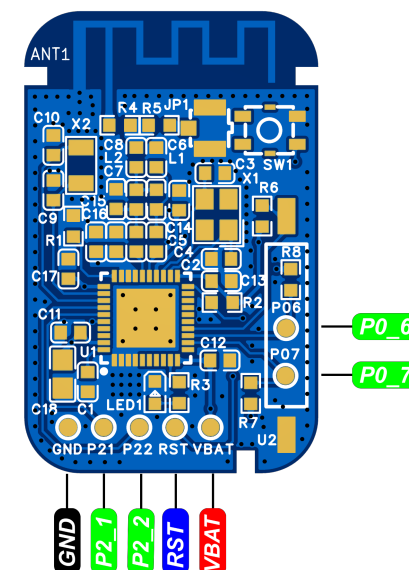
This project is a Zigbee End Device designed to operate as a battery-powered contact sensor (for doors, windows, or similar applications). It is based on the Texas Instruments CC2530 Zigbee SoC and can be flashed with the PTVO firmware to provide full Zigbee functionality.

Features

- Zigbee 1.2 compatible (using PTVO firmware)
- Based on CC2530F256 microcontroller
- Powered by a CR2032 coin cell
- Integrated reed switch for contact detection
- LED indicator for status or pairing feedback
- Reset button for firmware or network reset
- Compact design suitable for enclosure integration

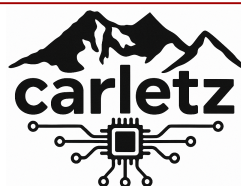
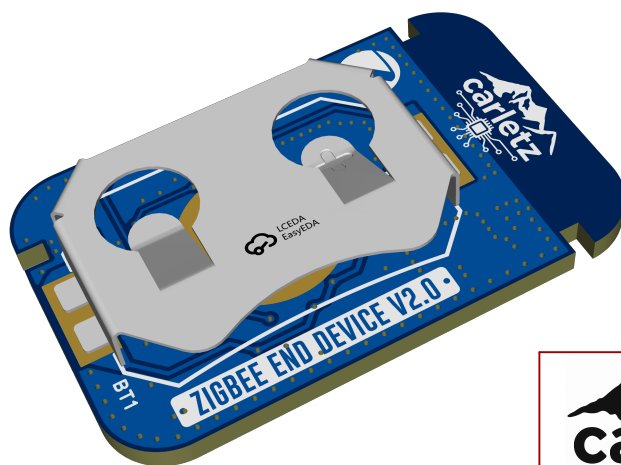
Connections:

CC2530 Pin	CC Debugger Pin
P2_2 (Debug Data)	DD
P2_1 (Debug Clock)	DC
GND	GND
VBAT	VCC (3.3 V)
RST	RESET



CC2530 Link to datasheet:

<https://www.lcsc.com/datasheet/C9120.pdf>

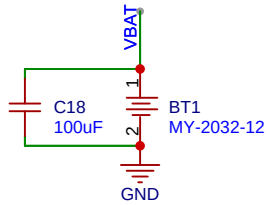


Zigbee end device V21

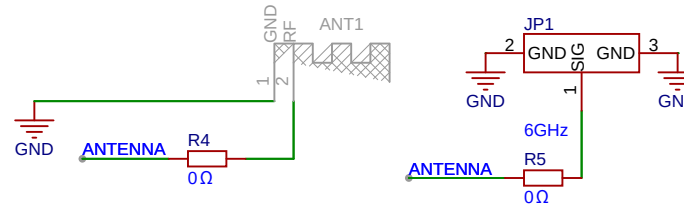
Version	Size	Page 1 Total 2
1	A4	EasyEDA

Create at	2026-01-10	Drawn	carletz
Update at	2026-01-10	Reviewed	carletz
Schematic	End Device (contact)		
Board	Zigbee End Device	Page	Description

Battery Power



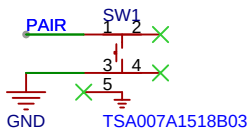
Antenna



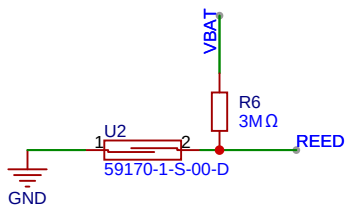
Antenna Note:

The R4 and R5 0Ω resistor are placed for selecting one or the other type of antenna.
The JP1 connector foresees the possibility to connect an external antenna.
Please choose the correct one for 2.4GHz.

Reset



Reed Switch



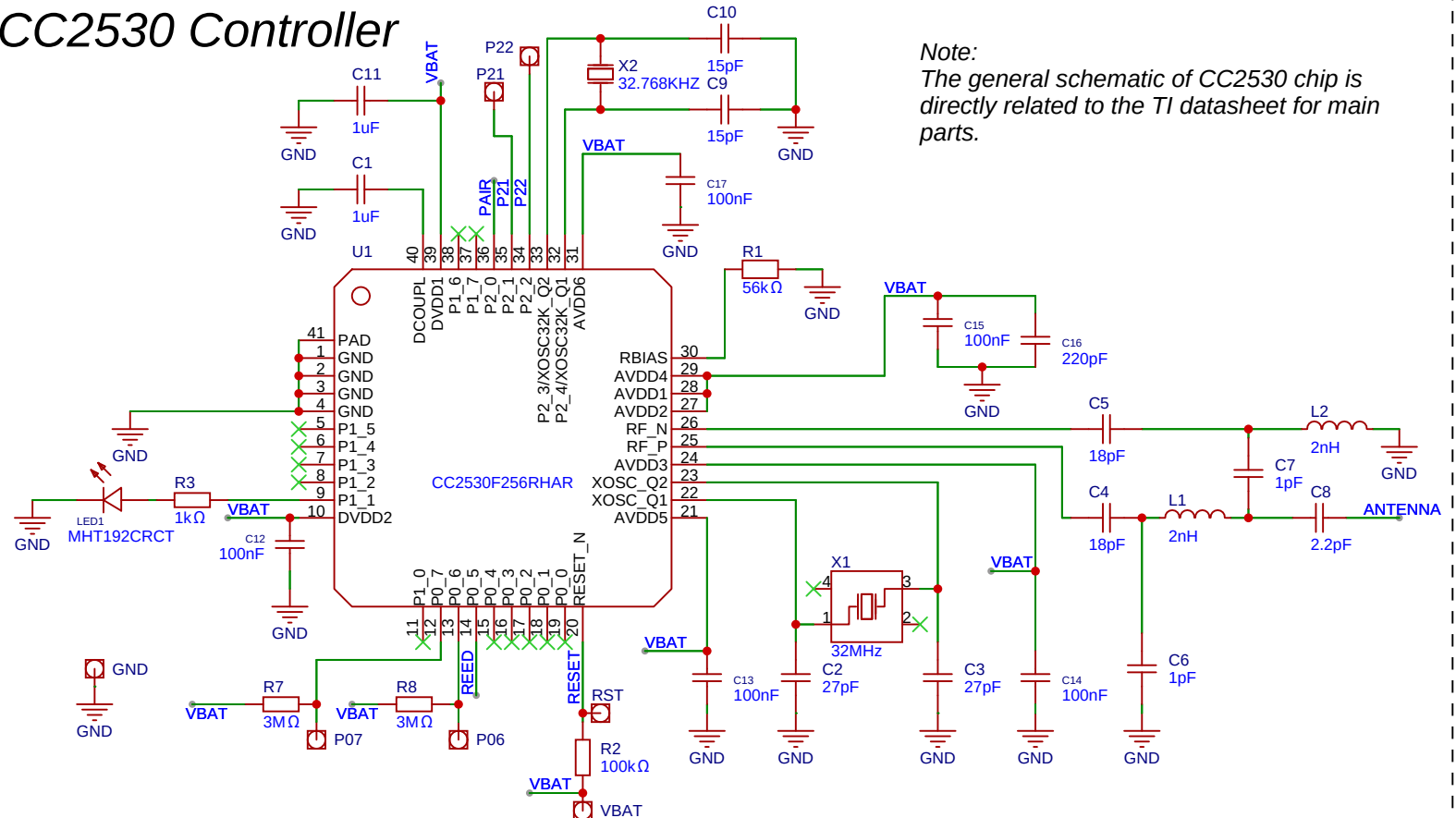
Reed Switch Note:

The R6, R7 and R8 resistor are foreseen in order to avoid battery drainage during reed readings.

The value can be chosen between 2 and 5 MΩ.

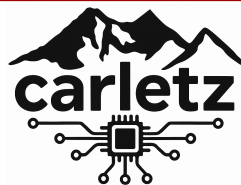
The connections P06 and P07 are for future improvements.

CC2530 Controller



Note:

The general schematic of CC2530 chip is directly related to the TI datasheet for main parts.



Zigbee end device V21

Version	Size	Page 2 Total 2
2	A4	EasyEDA

Create at	2026-01-09	Drawn	carletz
Update at	2026-01-10	Reviewed	carletz
Schematic	End Device (contact)		
Board	Zigbee End Device	Page	Schematic